

USHARAL, Kazakhstan

WMO: 367290

Lat: 46.167N

Lon: 80.933E

Elev: 397

StdP: 96.65

Time Zone: 6.00 (E06)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-22.5	-28.9	0.3	-26.1	-25.4	0.4	-22.2	12.6	-9.0	10.2	-7.5	1.1	70	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	34.6	19.4	33.1	18.9	31.4	18.3	20.9	31.5	20.1	30.0	19.3	29.0	2.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.2	24.8	16.5	12.3	24.1	15.6	11.6	23.9	62.4	31.7	59.4	30.1	56.7	29.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.1	5.9	-30.1	37.7	3.8	1.4	-32.8	38.7	-35.0	39.5	-37.2	40.3	-39.9	41.4		
(5)			-30.3	23.0	3.7	1.1	-33.0	23.8	-35.1	24.4	-37.2	25.0	-39.9	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-11.0	-10.2	-1.1	9.5	16.1	21.4	24.0	22.2	16.3	8.0	-0.2	-5.9		
	DBStd	13.15	5.55	5.92	6.25	4.73	4.33	3.36	2.62	3.13	4.18	4.88	5.97	5.21		
	HDD10.0	2534	651	567	347	64	7	0	0	0	4	94	307	494		
	HDD18.3	4403	909	800	603	266	94	11	1	5	85	321	556	752		
	CDD10.0	1624	0	0	2	49	194	342	434	377	193	33	1	0		
	CDD18.3	452	0	0	0	1	24	102	177	124	23	1	0	0		
	CDH23.3	5556	0	0	0	29	355	1139	2030	1564	423	15	0	0		
	CDH26.7	2286	0	0	0	3	97	441	935	679	128	2	0	0		
(14) Wind	WSAvg	1.9	1.7	1.8	2.2	2.5	2.1	1.7	1.6	1.6	1.6	1.8	2.0	1.8		
(15) Precipitation	PrecAvg	285	24	18	21	32	26	25	23	12	11	25	35	34		
	PrecMax	405	79	41	88	126	93	79	64	39	32	64	81	91		
	PrecMin	134	3	0	1	0	0	4	1	0	0	0	1	2		
	PrecStd	65	15	10	19	29	22	18	18	10	8	18	21	21		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.4	3.4	19.1	26.9	31.5	35.0	37.2	36.1	32.7	25.2	15.4	7.2		
		MCWB	2.0	2.1	10.6	14.5	16.7	19.8	20.4	19.5	18.5	14.2	9.8	4.1		
	2%	DB	1.0	1.5	14.5	23.7	29.4	32.7	35.0	34.3	30.1	21.6	12.1	4.0		
		MCWB	-0.3	0.2	8.4	12.8	16.2	18.9	19.6	19.0	16.6	12.3	7.8	2.4		
	5%	DB	-1.1	-0.3	10.4	21.3	27.4	31.0	33.4	32.6	27.8	19.3	9.8	2.3		
		MCWB	-2.3	-1.4	6.0	12.1	15.7	18.2	19.2	18.4	15.4	11.4	6.6	1.2		
	10%	DB	-3.2	-1.7	7.3	18.7	25.2	29.4	31.6	30.8	25.7	16.7	7.2	0.9		
		MCWB	-4.3	-2.8	4.2	10.8	14.5	17.6	18.8	17.9	14.7	9.8	4.7	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.5	11.0	16.2	18.6	21.3	22.6	21.8	19.1	15.0	10.5	4.7		
		MCDB	3.2	3.4	18.6	21.6	28.6	31.8	31.6	32.3	31.4	22.5	14.0	6.6		
	2%	WB	-0.1	0.4	8.5	13.9	17.3	20.1	21.3	20.6	17.2	12.9	8.5	2.7		
		MCDB	0.9	1.3	13.6	21.7	26.9	30.3	31.5	31.2	28.2	20.7	11.7	3.8		
	5%	WB	-2.2	-1.4	6.4	12.6	16.5	19.2	20.3	19.5	16.1	11.6	6.6	1.3		
		MCDB	-1.2	-0.3	10.6	19.7	25.4	29.0	30.1	30.0	26.7	18.8	9.5	2.2		
	10%	WB	-4.2	-2.8	4.3	11.2	15.4	18.3	19.6	18.5	15.0	10.1	4.8	-0.2		
		MCDB	-3.3	-1.7	7.3	18.0	23.7	27.6	29.0	28.5	24.4	16.0	7.0	0.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	9.1	12.9	13.3	13.0	13.3	14.6	15.0	13.0	9.2	7.9		
		MCDBR	10.4	9.1	13.8	17.7	17.5	15.4	16.6	17.5	18.8	17.0	13.0	9.0		
	5% WB	MCWBR	9.4	8.1	8.2	8.9	7.6	6.3	6.1	7.1	8.8	9.0	9.0	7.5		
		MCWBR	10.4	8.5	13.1	14.5	15.0	14.0	13.9	15.3	18.3	15.7	11.9	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.358	0.403	0.442	0.427	0.433	0.435	0.399	0.372	0.374	0.356	0.320		
	taud		2.069	2.042	2.072	2.045	2.149	2.197	2.210	2.289	2.314	2.203	2.143	2.089		
	Ebn at Noon		737	784	808	813	841	837	829	846	838	763	688	685		
	Edh at Noon		98	124	140	157	147	141	137	122	109	105	91	86		
(44) All-Sky Solar Radiation	RadAvg		1.80	2.77	3.91	5.21	6.58	6.85	6.58	6.10	4.84	3.02	1.77	1.38		
	RadStd		0.12	0.14	0.26	0.35	0.30	0.31	0.40	0.32	0.28	0.26	0.15	0.11		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page